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p01-01. The first number is the x-coordinate (horizontal movement); the second is the y-coordinate (vertical movement). For $(4, 2)$, move 4 right and 2 up.

p01-02. Correct plot: $A(2, 5)$ and $B(5, 1)$.

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p02-01. $(3, 6)$: 3 units right, then 6 units up.

p02-02. x-coordinate 7; y-coordinate 4; move 7 right and 4 up.

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p03-01. The vertices form a square: each side is 4 units.

p03-02. $H(2, 6)$ and $I(4, 6)$ are tied for highest because both have y-coordinate 6. I and J share x-coordinate 4.

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p04-01. $(2 + 4, 3) = (6, 3)$. Only x changes because the move is horizontal.

p04-02. $(6, 5 - 3) = (6, 2)$. Only y changes; moving down decreases y.

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p05-01. $M' = (1 + 3, 4 + 2) = (4, 6)$.

p05-02. $N' = (7 - 5, 2 - 1) = (2, 1)$.

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p06-01. $P'(3, 1)$, $Q'(5, 1)$, $R'(4, 4)$.

p06-02. $(1, 5)$, $(5, 5)$, $(5, 7)$, $(1, 7)$.

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p07-01. $A' = (3, 3)$. The original point is 1 unit above $y = 4$, so its reflection is 1 unit below that line. The x-coordinate stays 3.

p07-02. $B'(1, 6)$, $C'(4, 6)$, $D'(2, 2)$. The y-coordinates 2 and 6 are each 2 units from $y = 4$, on opposite sides; each x-coordinate stays the same.

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p08-01. $E' = (2, 2)$. The x-coordinate changes from 6 to 2: the point moves from 2 units right of $x = 4$ to 2 units left of it. The y-coordinate stays 2.

p08-02. $(7, 1)$, $(4, 1)$, $(4, 3)$, $(7, 3)$, in the original vertex order. Points at x-coordinate 1 reflect to x-coordinate 7, each 3 units from $x = 4$. Points on $x = 4$ stay in place.

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p09-01. It stays $(4, 4)$ because the point is on the mirror line $x = 4$ and is zero units from that line.

p09-02. The blank is 5: $(5, 3)$ is 1 unit below $y = 4$, so its reflection $(5, 5)$ is 1 unit above the line.

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p10-01. First quarter-turn clockwise: right. Second: down.

p10-02. Label the sketch $A'(1, 3)$, $B'(1, 1)$, $C'(3, 1)$, and $D'(3, 3)$, and connect them in order back to A' . Relative to center $(2, 2)$, the lower-left vertex moves to the upper-left, the lower-right to the lower-left, the upper-right to the lower-right, and the upper-left to the upper-right. The square covers the same region, but its vertex labels move.

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p11-01. $G = (6, 4)$. Two units right of $(4, 4)$ rotated clockwise becomes two units below: $G' = (4, 2)$.

p11-02. $H = (5, 8)$. Three units above $(5, 5)$ rotated counterclockwise becomes three units left: $H' = (2, 5)$.

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p12-01. $7 - 2 = 5$ units.

p12-02. $6 - 1 = 5$ units.

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p13-01. Width $6 - 1 = 5$, height $4 - 1 = 3$; perimeter $2(5 + 3) = 16$ units.

p13-02. Side length $5 - 2 = 3$; perimeter $4 \times 3 = 12$ units.

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p14-01. Width $8 - 2 = 6$, height $4 - 1 = 3$; area $6 \times 3 = 18$ square units.

p14-02. Side length $7 - 3 = 4$; area $4 \times 4 = 16$ square units.

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p15-01. Yes. Congruent figures have the same shape and size; both rectangles have corresponding sides 4 and 7.

p15-02. A translation moves every point the same distance and direction, so size and shape stay the same. The figures are congruent.

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p16-01. A translation slides a figure; every point moves the same distance in the same direction.

p16-02. The vertical line is $x = 5$, halfway between x-coordinates 4 and 6. Each point is 1 unit from the line on an opposite side, and both have y-coordinate 2.

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p17-01. Translate: $(2, 3) \rightarrow (4, 3)$. Reflect across $x = 5$: $(4, 3) \rightarrow (6, 3)$, moving from 1 unit left of the line to 1 unit right. Final coordinate: $(6, 3)$.

p17-02. Reflect across $y = 3$: $(5, 1) \rightarrow (5, 5)$, from 2 units below the line to 2 units above. Translate: $(5 - 1, 5 + 2) = (4, 7)$.

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p18-01. Same x-coordinate, so vertical distance is $5 - 1 = 4$ blocks.

p18-02. East increases x: $3 + 4 = 7$. South decreases y: $4 - 3 = 1$. Stop at $(7, 1)$.

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p19-01. Lower-left $(1, 1)$; vertices are $(1, 1)$, $(4, 1)$, $(4, 3)$, $(1, 3)$.

p19-02. Horizontal side length $4 - 1 = 3$; vertical side length $5 - 2 = 3$.

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p20-01. Translate right: $(1, 2), (5, 2), (5, 5), (1, 5) \rightarrow (3, 2), (7, 2), (7, 5), (3, 5)$. Reflect across $y = 4$: y-coordinate 2 becomes 6, and 5 becomes 3, keeping equal distances on opposite sides of the line. The final vertices, in order, are $(3, 6), (7, 6), (7, 3), (3, 3)$.

p20-02. Sample answer: both transformations preserve size and shape. A translation slides every point the same distance in the same direction; a reflection flips a figure across a mirror line. Example translation 2 units right: $(1, 2) \rightarrow (3, 2)$. Example reflection across $x = 4$: $(1, 2) \rightarrow (7, 2)$, since both points are 3 units from that line. Other correct similarities, differences, and examples are acceptable.